Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Processing (01CT1513)	**Aim : Design FIR filter with windowing method.**	
Experiment : 6	**Date: 23/10/2026**	**Enrollment No: 92400133125**

Github : https://github.com/mankumarbhalodiya/Digital-Signal-and-Image-Processing

AIM: Design FIR filter with windowing method.

Code :

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import firwin, freqz
N = 31
Fs = 10000
low = firwin(N, 3000, window='hann', fs=Fs)
high = firwin(N, 3000, window='hann', pass_zero='highpass', fs=Fs)
band = firwin(N, [3000, 4500], window='hann', pass_zero='bandpass', fs=Fs)
stop = firwin(N, [3000, 4500], window='hann', pass_zero='bandstop', fs=Fs)
filters = [low, high, band, stop]
titles = ['Low Pass', 'High Pass', 'Band Pass', 'Band Stop']
plt.figure(figsize=(10, 8))
for i in range(4):
    f, H = freqz(filters[i], worN=2048, fs=Fs)
    plt.subplot(2, 2, i + 1)
    plt.plot(f, 20 * np.log10(np.maximum(abs(H), 1e-10)))
    plt.title(titles[i])1
    plt.xlabel('Frequency (Hz)')
```

```
    plt.ylabel('Magnitude (dB)')
    plt.grid()
plt.tight_layout()
plt.show()
plt.figure(figsize=(10, 8))
for i in range(4):
    plt.subplot(2, 2, i + 1)
    plt.stem(filters[i])
    plt.title(titles[i])
    plt.xlabel('Sample')
    plt.ylabel('Amplitude')
    plt.grid()
plt.tight_layout()
plt.show()
plt.figure(figsize=(10, 8))
for i in range(4):
    f, H = freqz(filters[i], worN=2048, fs=Fs)
    plt.subplot(2, 2, i + 1)
    plt.plot(f, np.unwrap(np.angle(H)))
    plt.title(titles[i])
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Phase (Radians)')
```

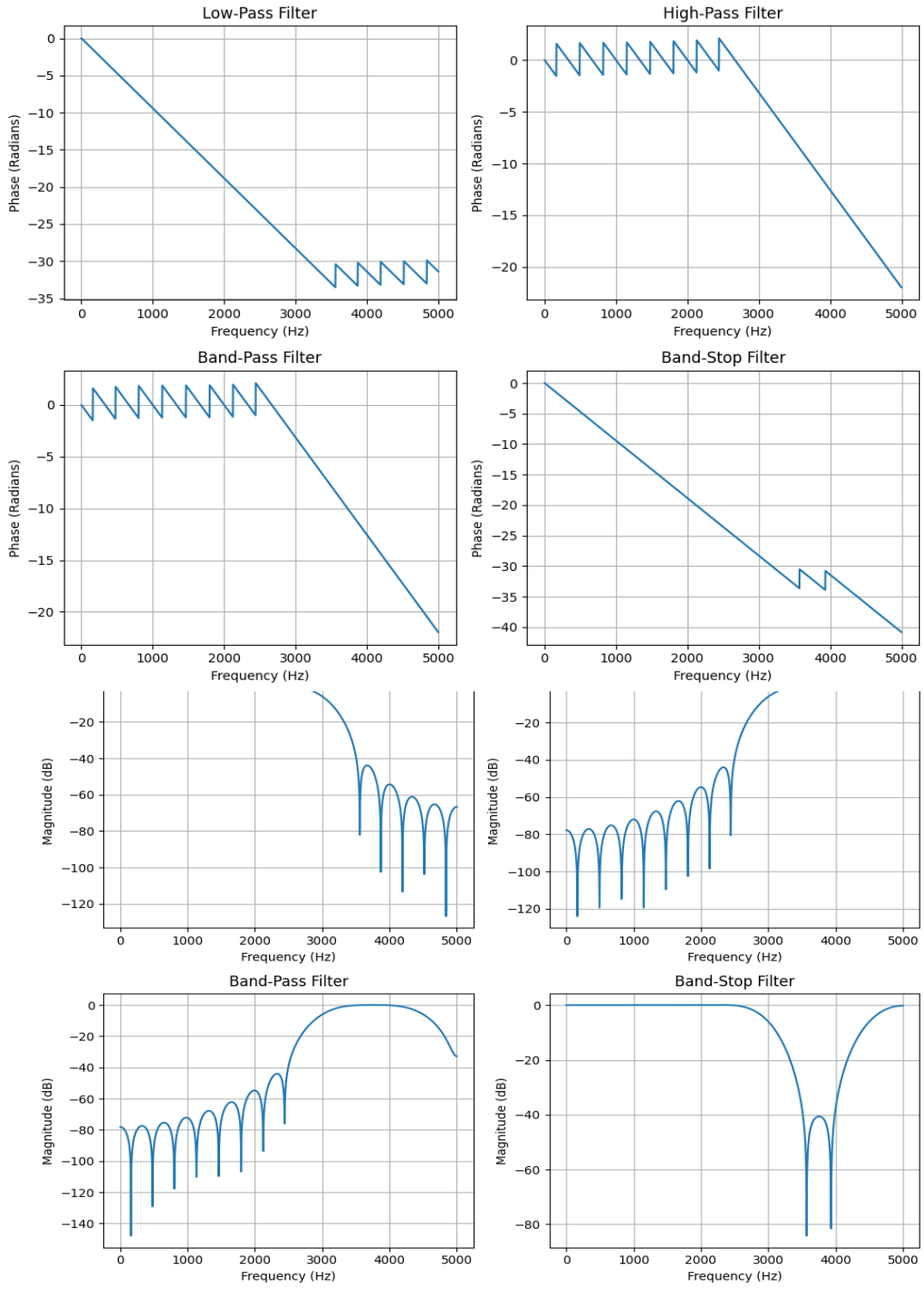
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```
    plt.grid()

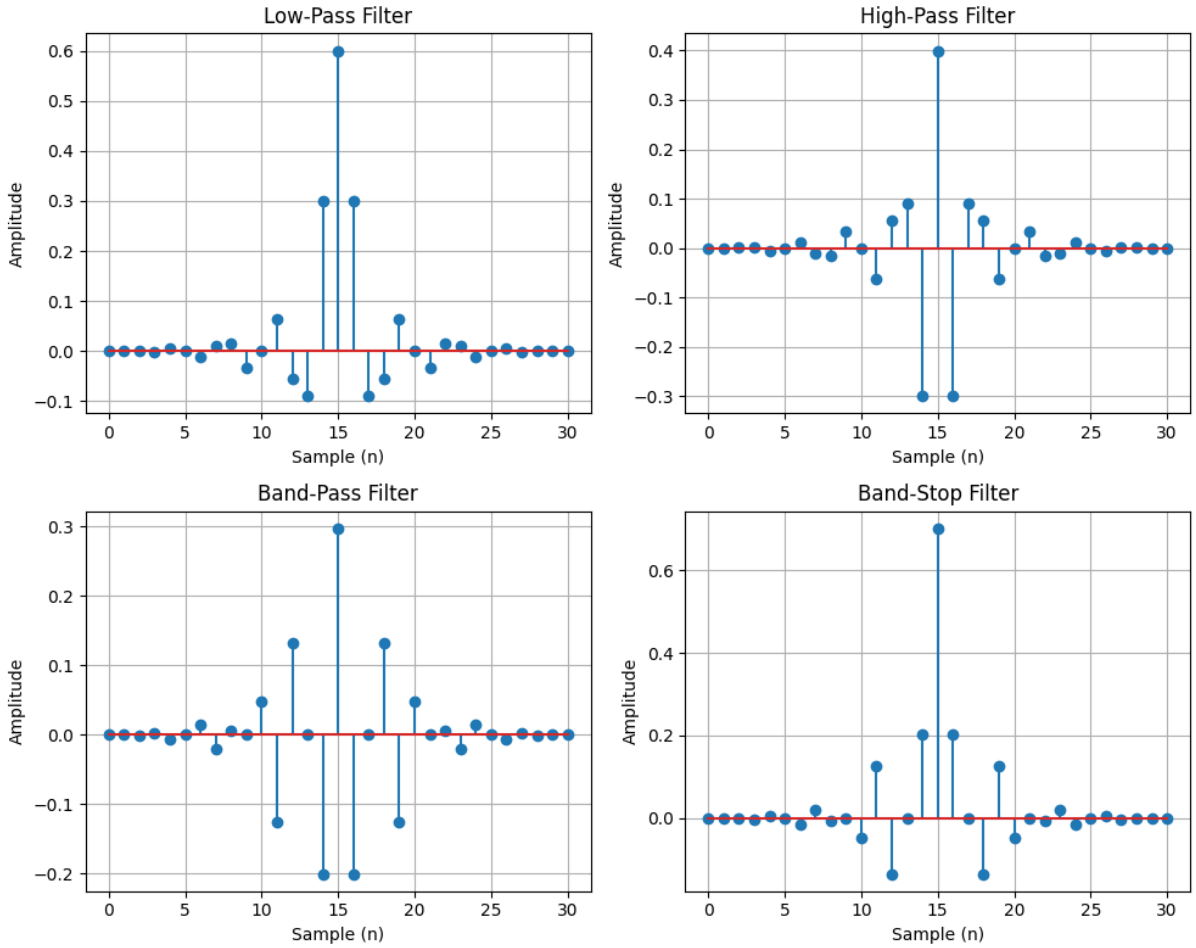
plt.tight_layout()

plt.show()
```

Output :



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Music Anylysis Code :

```
import numpy as np

import matplotlib.pyplot as plt

import librosa

from scipy.io import wavfile

from scipy.signal import firwin, lfilter
```

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```
song, Fs =
librosa.load("Todh___Prince_Narula___Munawar__Official_Video____Jaymeet___R
ony_Ajnali___Gill_Machhrai(128k).mp3", sr=None)

N = 31

low = firwin(N, 3000, window='hann', fs=Fs)

high = firwin(N, 3000, window='hann', pass_zero='highpass', fs=Fs)

band = firwin(N, [3000, 6000], window='hann', pass_zero='bandpass', fs=Fs)

stop = firwin(N, [3000, 6000], window='hann', pass_zero='bandstop', fs=Fs)

filters = [low, high, band, stop]

titles = ['Low Pass', 'High Pass', 'Band Pass', 'Band Stop']

files = ['low_pass.wav', 'high_pass.wav', 'band_pass.wav', 'band_stop.wav']

plt.figure(figsize=(10, 8))

for i in range(4):

    output = lfilter(filters[i], 1, song)

    plt.subplot(4, 1, i + 1)

    plt.plot(np.arange(len(output)) / Fs, output)

    plt.title(titles[i])

    plt.xlabel('Time (seconds)')

    plt.ylabel('Amplitude')

    plt.grid()

    wavfile.write(files[i], Fs, (output / (np.max(np.abs(output)) + 1e-10) *
32767).astype(np.int16))
```

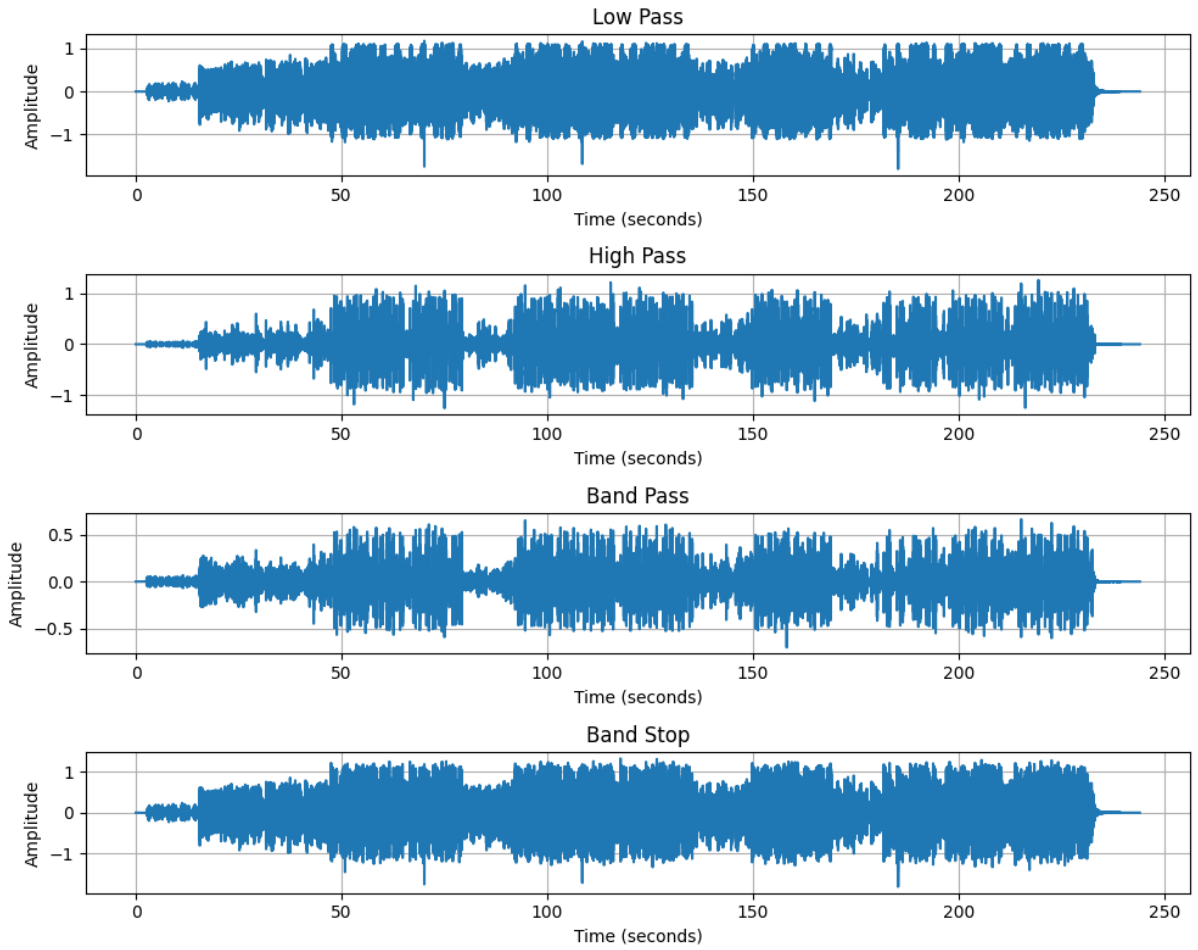
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```
plt.tight_layout()

plt.show()

print("All filtered songs saved!")
```

Output :



Conclusion : The four FIR filters were successfully applied to the song using Python. The low-pass filter reduces high-frequency sounds, while the high-pass filter reduces low-frequency sounds. The band-pass filter keeps sounds between 3 kHz and 6 kHz, and the band-stop filter removes sounds in that

range. After filtering, all four audio files were saved and their waveforms were displayed for comparison.